

CHAPTER 2 : SIMPLE LINEAR REGRESSION

Introduction to Regression

Q1) Prove the following:

$$\forall v(x): \mathbb{R} \rightarrow \mathbb{R}, \text{Cov}(Y - E[Y|X], v(x)) = 0$$

Approach #1

$$\begin{aligned} \text{Cov}(Y - E[Y|X], v(x)) &= E[(Y - E[Y|X]) \cdot v(x)] - E[Y - E[Y|X]] E[v(x)] \\ &= \{E[Y \cdot v(x)] - E[E[Y \cdot v(x)|X]]\} - \{E[Y] - E[E[Y|X]]\} E[v(x)] \\ &= \{E[Y \cdot v(x)] - E[Y \cdot v(x)]\} \\ &= 0 \end{aligned}$$

Approach #2

$$\begin{aligned} \text{Cov}(Y - E[Y|X], v(x)) &= E[(Y - E[Y|X]) v(x)] \\ &= E[E[(Y - E[Y|X]) v(x) | X]] \\ &= E[v(x) E[Y|X] - v(x) E[E[Y|X] | X]] \\ &= E[v(x) E[Y|X] - v(x) E[Y|X]] \\ &= 0 \end{aligned}$$

Q2) Prove the following:

$$\underset{u(x)}{\text{argmin}} E[(Y - u(x))^2] = E[Y|X]$$

$$E[(Y - u(x))^2] = E[(Y - E[Y|X] + E[Y|X] - u(x))^2]$$

$$\begin{aligned} &= E[(Y - E[Y|X])^2] + 2E[(Y - E[Y|X])(E[Y|X] - u(x))] + E[(E[Y|X] - u(x))^2] \\ &= E[(Y - E[Y|X])^2] + E[(E[Y|X] - u(x))^2] \\ &\geq E[(Y - E[Y|X])^2] \end{aligned}$$

$$\therefore \underset{u(x)}{\text{argmin}} E[(Y - u(x))^2] = E[Y|X]$$

Simple Linear Regression Model

Q3) What is an identifiable Model

$$\text{if } f(\cdot; \theta_0) = f(\cdot; \theta_1) \Leftrightarrow \theta_0 = \theta_1$$

$f(x; \theta)$ is an identifiable model

(or) An identifiable model is a model such that for two arbitrary regression models $f(\cdot; \theta_0)$ and $f(\cdot; \theta_1)$, if $f(\cdot; \theta_0) = f(\cdot; \theta_1)$, then $\theta_0 = \theta_1$

* Make a distinction between a regression model and a probability model

Q4) Show that the simple linear regression model is not identifiable if $S_{xx} = 0$

$$\text{Simple linear regression model: } y = \beta_0 + \beta_1 x + \varepsilon \quad \varepsilon \sim N(0, \sigma^2 I_n)$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 \quad \therefore \text{if } S_{xx} = 0 \Leftrightarrow x_1 = \dots = x_n = \bar{x}$$

Suppose that $\bar{x} = 1$

$$\text{First model: } (\beta_0, \beta_1) = (1, 0) \quad y_i = 1 + 0 \cdot \bar{x} + \varepsilon_i = 1 + 0 \cdot 1 + \varepsilon_i = 1 + \varepsilon_i$$

$$\text{Second model: } (\beta'_0, \beta'_1) = (0, 1) \quad y_i = 0 + 1 \cdot \bar{x} + \varepsilon_i = 0 + 1 \cdot 1 + \varepsilon_i = 1 + \varepsilon_i$$

$\therefore$ The model is the same but their respective parameters are different. Thus the model is NOT identifiable.

Method of Estimation

Q5) Show that $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased estimators of β_0 and β_1

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{(\beta_0, \beta_1) \in \mathbb{R}^2}{\text{argmin}} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = \underset{(\beta_0, \beta_1) \in \mathbb{R}^2}{\text{argmin}} L(\beta_0, \beta_1)$$

$$\begin{aligned} \frac{\partial L}{\partial \beta_0} &= -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) \\ \frac{\partial L}{\partial \beta_1} &= -2 \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) \end{aligned} \quad \begin{aligned} &\text{Sylvester's criterion:} \\ &+ \nabla^2 L(\beta_0, \beta_1) = \begin{pmatrix} 2n & 2\sum x_i \\ 2\sum x_i & 2\sum x_i^2 \end{pmatrix} \\ &= 2n \begin{pmatrix} 1 & \bar{x} \\ \bar{x} & \bar{x}^2 \end{pmatrix} \end{aligned}$$

$$\frac{\partial L}{\partial \beta_0} \Big|_{\beta_0 = \hat{\beta}_0} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \quad (A > 0 \text{ and } a_n > 0, \text{ and } b = c)$$

$$\therefore \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$\Downarrow$
critical points are global minima

$$\frac{\partial L}{\partial \beta_1} \Big|_{\beta_1 = \hat{\beta}_1} = -2 \sum_{i=1}^n x_i y_i - x_i \hat{\beta}_0 - \hat{\beta}_1 x_i^2 = 0$$

$$= \hat{\beta}_1 = \frac{\bar{xy} - \bar{x} \hat{\beta}_0}{\bar{x}^2}, \quad \bar{xy} = \frac{1}{n} \sum_{i=1}^n x_i y_i, \quad \bar{x}^2 = \frac{1}{n} \sum_{i=1}^n x_i^2$$

insert $\hat{\beta}_0$ into $\hat{\beta}_1$

$$\begin{aligned} \hat{\beta}_1 &= \frac{\bar{xy} - \bar{x}(\bar{y} - \hat{\beta}_1 \bar{x})}{\bar{x}^2} \Leftrightarrow \bar{x}^2 \hat{\beta}_1 = \bar{xy} - \bar{x} \bar{y} + \hat{\beta}_1 \bar{x}^2 \\ &\Leftrightarrow (\bar{x}^2 - \bar{x}^2) \hat{\beta}_1 = \bar{xy} - \bar{x} \bar{y} \\ &\therefore \hat{\beta}_1 = (\bar{xy} - \bar{x} \bar{y}) / (\bar{x}^2 - \bar{x}^2) \\ &= S_{xy} / S_{xx} \end{aligned}$$

$$\text{Let } S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\text{insert } \hat{\beta}_1 \text{ into } \hat{\beta}_0: \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \bar{y} - \frac{\sum xy}{\sum xx} \bar{x}$$

$$E[\hat{\beta}_1] = E\left[\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}\right]$$

$$+ E[y_i] = \beta_0 + \beta_1 x_i, E[\bar{y}] = \frac{1}{n} \sum_{i=1}^n E[y_i] = \beta_0 + \beta_1 \bar{x}$$

$$\begin{aligned} E\left[\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum xx}\right] &= \frac{\sum (x_i - \bar{x}) E[y_i - \bar{y}]}{\sum xx} \\ &= \frac{\sum (x_i - \bar{x})(\beta_1 x_i + \beta_1 \bar{x})}{\sum xx} \\ &= \beta_1 \frac{\sum (x_i - \bar{x})^2}{\sum xx} = \beta_1 \end{aligned}$$

$$E[\hat{\beta}_0] = E[\bar{y} - \bar{x} \hat{\beta}_1] = \beta_0 + \beta_1 \bar{x} - \bar{x} \beta_1 = \beta_0$$

$\therefore \hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased estimators.

Q6) Show that $\text{var}(\hat{\beta}_1)$ and $\text{var}(\hat{\beta}_0)$ are $\frac{\sigma^2}{\sum xx}$ and $(\frac{1}{n} + \frac{\bar{x}^2}{\sum xx}) \sigma^2$ respectively

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum xy}{\sum xx}, \text{var}(\hat{\beta}_1) = \text{var}\left(\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum xx}\right) \\ &= \frac{1}{\sum xx^2} \text{var}\left(\sum_{i=1}^n (x_i - \bar{x}) y_i\right) \\ &= \frac{\sum (x_i - \bar{x})^2}{\sum xx^2} \text{var}(y_i) = \frac{\sigma^2}{\sum xx} \end{aligned}$$

$$\begin{aligned} \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}, \text{var}(\bar{y} - \hat{\beta}_1 \bar{x}) = \text{var}(\bar{y}) - 2 \text{cov}(\bar{y}, \hat{\beta}_1 \bar{x}) + \text{var}(\hat{\beta}_1 \bar{x}) \\ &= \frac{\sigma^2}{n} - 0 + \frac{\bar{x}^2}{\sum xx} \sigma^2 = \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum xx}\right) \sigma^2 \\ \therefore \text{cov}(\bar{y}, \hat{\beta}_1) &= 0 \end{aligned}$$

Q7) Show that $\text{cov}(\bar{y}, \hat{\beta}_1) = 0$

$$\begin{aligned} \text{cov}(\bar{y}, \hat{\beta}_1) &= \text{cov}\left(\frac{1}{n} \sum_{i=1}^n y_i, \sum_{i=1}^n a_i y_i\right) \quad a_j = \frac{(x_j - \bar{x})}{\sum xx} \\ &= \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^n a_i \text{cov}(y_i, y_j) \\ &= \frac{1}{n} \sum_{i=1}^n a_i \text{cov}(y_i, y_i) = \frac{1}{n} \sigma^2 \sum_{i=1}^n a_i = 0 \quad \because \sum_{i=1}^n a_i = 0 \end{aligned}$$

Q8) Show that $\hat{\beta}_1$ and $\hat{\beta}_0$ are the minimum critical points for $L(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$

Question is answered in Q5)

Bonus Question: How do we know if a Hessian Matrix is positive definite?

Let $H = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. If $\det(H) > 0$ and $a_{11} > 0$, H is a positive definite matrix.

Q9) Prove the following:

$$\begin{aligned} \text{(a)} \quad E[\hat{\mu}_1] &= \mu_1 & \text{(e)} \quad \text{Var}(\hat{\mu}_1) &= \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x} - \bar{x})^2}{\sum xx} \right) \\ \text{(b)} \quad E[\hat{y}_2] &= E[y_2] & \text{(f)} \quad \text{Var}(\hat{y}_2) &= \sigma^2 \left(\frac{1}{n} + \frac{(x_2 - \bar{x})^2}{\sum xx} \right) \\ \text{(c)} \quad E[\hat{\epsilon}_1] &= E[\epsilon_1] & \text{(g)} \quad \text{Var}(\hat{\epsilon}_1) &= \sigma^2 \left(1 - \frac{1}{n} - \frac{(x_1 - \bar{x})^2}{\sum xx} \right) \\ \text{(d)} \quad E[\hat{\sigma}^2] &= \sigma^2 \end{aligned}$$

$$\text{(a)} \quad \hat{\mu}_1 = \hat{y}_1 = \hat{\beta}_0 + \hat{\beta}_1 x_1, E[\hat{\mu}_1] = \beta_0 + \beta_1 x_1 = E[\beta_0 + \beta_1 x_1 + \epsilon_1] = \mu_1$$

$$\begin{aligned} \text{(b)} \quad \hat{y}_2 &= \hat{\beta}_0 + \hat{\beta}_1 x_2, y_2 = \beta_0 + \beta_1 x_2 + \epsilon_2, \epsilon_2 \sim N(0, \sigma^2) \\ E[\hat{y}_2] &= \beta_0 + \beta_1 x_2 = E[\beta_0 + \beta_1 x_2 + \epsilon_2] = E[y_2] \end{aligned}$$

$$\text{(c)} \quad \hat{\epsilon}_1 = y_1 - \hat{y}_1, E[\hat{\epsilon}_1] = E[y_1] - E[\hat{y}_1] = 0 = E[\epsilon_1]$$

$$\begin{aligned} \text{(d)} \quad \hat{\sigma}^2 &= \frac{1}{n-2} \sum_{i=1}^n \hat{\epsilon}_i^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ E[\hat{\sigma}^2] &= \frac{1}{n-2} \sum_{i=1}^n E[\hat{\epsilon}_i^2] = \frac{1}{n-2} \sum_{i=1}^n \{E[\epsilon_i^2] - E[\epsilon_i]^2\} \\ &= \frac{1}{n-2} \sum_{i=1}^n \text{Var}(\epsilon_i), \text{Var}(\epsilon_i) = \sigma^2 \Rightarrow \frac{1}{n-2} \sum_{i=1}^n \left(1 - \frac{1}{n} - \frac{(x_i - \bar{x})^2}{\sum xx}\right) \sigma^2 = \sigma^2 \end{aligned}$$

$$\begin{aligned} \text{(e)} \quad \text{Var}(\hat{\mu}_1) &= \text{Var}(\hat{y}_1) = \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_1) = \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_1) \\ &= \text{Var}(\bar{y}) + (\bar{x} - x_1)^2 \text{Var}(\hat{\beta}_1) \\ &= \frac{\sigma^2}{n} + (\bar{x} - x_1)^2 \frac{\sigma^2}{\sum xx} = \sigma^2 \left(\frac{1}{n} + \frac{(\bar{x} - x_1)^2}{\sum xx} \right) \end{aligned}$$

$$\begin{aligned} \text{(f)} \quad \text{Var}(\hat{y}_2) &= \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_2) = \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_2) \\ &= \text{Var}(\bar{y}) + \text{Var}((x_2 - \bar{x}) \hat{\beta}_1) = \sigma^2 \left(\frac{1}{n} + \frac{(x_2 - \bar{x})^2}{\sum xx} \right) \end{aligned}$$

$$\begin{aligned} \text{(g)} \quad \text{Var}(\hat{\epsilon}_1) &= \text{Var}(y_1 - \hat{y}_1) = \text{Var}(y_1) - 2 \text{cov}(y_1, \hat{y}_1) + \text{Var}(\hat{y}_1) \\ &= \sigma^2 - 2 \text{cov}(\hat{y}_1 + \hat{\epsilon}_1, \hat{y}_1) + \text{Var}(\hat{y}_1) \\ &= \sigma^2 - \text{Var}(\hat{y}_1) \\ &= \sigma^2 \left(1 - \frac{1}{n} - \frac{(x_1 - \bar{x})^2}{\sum xx} \right) \\ &= \text{cov}(y_1 + \hat{\epsilon}_1, y_1) \\ &= \text{cov}(\hat{y}_1, y_1) + \text{cov}(\hat{\epsilon}_1, y_1) \\ &= \text{Var}(\hat{y}_1) + 0 \end{aligned}$$

Analysis Of Variance

Q10) Prove the following:

$$\begin{aligned} \sum_{i=1}^n (y_i - \bar{y})^2 &= \sum_{i=1}^n \{ (y_i - \hat{y}_i)^2 + 2(y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) + (\hat{y}_i - \bar{y})^2 \} \\ &= \sum (y_i - \hat{y}_i)^2 + \sum (\hat{y}_i - \bar{y})^2 \quad \because \sum (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) = 0 \end{aligned}$$

$$\begin{aligned} + \sum (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) &= \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y}) \\ &= \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(\hat{\beta}_0 - \bar{y} + \hat{\beta}_1 x_i) \\ &= (\hat{\beta}_0 - \bar{y}) \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) + \hat{\beta}_1 \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i \\ &= (\hat{\beta}_0 - \bar{y}) \left[\frac{\partial}{\partial \beta_0} L(\beta_0, \beta_1) \right]_{\beta=\hat{\beta}} + \hat{\beta}_1 \left[\frac{\partial}{\partial \beta_1} L(\beta_0, \beta_1) \right]_{\beta=\hat{\beta}} \\ &= (\hat{\beta}_0 - \bar{y}) \cdot 0 + \hat{\beta}_1 \cdot 0 = 0 \end{aligned}$$

Q11) Prove the following:

Given that $R^2 = \frac{SSR}{SST}$, prove the following:

(a) $R^2 = 0 \Leftrightarrow \hat{\beta}_1 = 0$

(b) $R^2 = 1 \Leftrightarrow y_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \quad (i=1, \dots, n)$

(c) $R^2 = \frac{S_{xy}}{S_{xx} S_{yy}} \quad (= \text{Square of sample correlation})$

(Solution)

$R^2 = SSR/SST$ (SSR = Sum Square Regression, SST = Sum Square Total)

$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$ "Sum of Squares due to Regression" "Total Sum of Square"

$SST = \sum_{i=1}^n (y_i - \bar{y})^2$

$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$

(a) $R^2 = 0 \Leftrightarrow SSR = 0 \Leftrightarrow \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = 0 \Leftrightarrow \hat{y}_i = \bar{y} \quad \forall i = (1, \dots, n)$
 $\Leftrightarrow \hat{\beta}_0 + \hat{\beta}_1 x_i = \bar{y} \quad \forall i = (1, \dots, n)$
 $\Leftrightarrow \hat{\beta}_0 = \bar{y} \text{ and } \hat{\beta}_1 = 0 \text{ or } x_1 = \dots = x_n$

However the supposition is that the model is identifiable.

i.e., $S_{xx} \neq 0 \therefore x_1 = \dots = x_n$ is eliminated.

$\therefore \hat{\beta}_0 = \bar{y} \text{ and } \hat{\beta}_1 = 0$

(b) $R^2 = 1 \Leftrightarrow SSR = SST \Leftrightarrow \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \bar{y})^2 \quad \forall i = (1, \dots, n)$

$\therefore \hat{y}_i = y_i \quad \forall i = (1, \dots, n)$

$\Leftrightarrow \hat{\beta}_0 + \hat{\beta}_1 y_i = y_i \quad \forall i = (1, \dots, n)$

(c) $R^2 = \frac{SSR}{SST} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y})^2}{S_{yy}} = \frac{\sum_{i=1}^n (\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i - \bar{y})^2}{S_{yy}} = \frac{\hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2}{S_{yy}} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = \left[\frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \right]^2$

$\frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} = \frac{\frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2 \frac{1}{n} \sum (y_i - \bar{y})^2}} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \rho$

Distribution of Estimator (Hypothesis Testing)

Q12) Prove Theorem 11

(a) $\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}})$

(b) $\frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-2)$

(c) $\hat{\beta}_1$ and $\hat{\sigma}^2$ are independent

* note: $SSR = \hat{\beta}_1 S_{xy}$, $SSE = (n-2)\hat{\sigma}^2$

(a) $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x}) y_i}{S_{xx}} \sim N(E[\hat{\beta}_1], \text{Var}(\hat{\beta}_1)) \therefore y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$

$\therefore \hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}})$

(b) $\frac{(n-2)\hat{\sigma}^2}{\sigma^2} = \frac{(n-2)}{\sigma^2} \sum_{i=1}^n \hat{e}_i^2 = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
 $= \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \frac{n-2}{\sigma^2} \sum_{i=1}^n (y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i)^2$
 $= \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \bar{y} - \hat{\beta}_1 (x_i - \bar{x}))^2$
 $= \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \bar{y} - \left[\frac{\sum (x_j - \bar{x}) y_j}{S_{xx}} \right] (x_i - \bar{x}))^2$

Let $A = \left(\frac{x_1 - \bar{x}}{S_{xx}}, \dots, \frac{x_n - \bar{x}}{S_{xx}} \right)^T \therefore \hat{\beta}_1 = A^T y$

$= \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \bar{y} - A^T y \cdot \frac{(x_i - \bar{x})}{S_{xx}} \cdot S_{xx})^2$
 $= \frac{1}{\sigma^2} \|y - \Pi_{\mathbb{1}} y - A A^T y\|^2$ * Notice that $S_{xx} = (A^T A)^{-1}$

$= \frac{1}{\sigma^2} \|y - \Pi_{\mathbb{1}} y - A(A^T A)^{-1} A^T y\|^2$

$= \frac{1}{\sigma^2} \|y - \Pi_{\mathbb{1}} y - \Pi_A y\|^2$

$\Pi_{\mathbb{1}}$ and Π_A are orthogonal because $\mathbb{1}^T A = \frac{\sum (x_i - \bar{x})}{S_{xx}} = 0$

$\therefore \Pi_{\mathbb{1}} + \Pi_A$ is a projection

$\Leftrightarrow I_n - \Pi_{\mathbb{1}} - \Pi_A$ is a projection

$\Leftrightarrow (I_n - \Pi_{\mathbb{1}} - \Pi_A)^2 = I_n - \Pi_{\mathbb{1}} - \Pi_A$ is idempotent

$\therefore \frac{1}{\sigma^2} \|y - \Pi_{\mathbb{1}} y - \Pi_A y\|^2 = \frac{1}{\sigma^2} y^T (I_n - \Pi_{\mathbb{1}} - \Pi_A) y \sim \chi^2(n-2; \delta)$

$\therefore A \Sigma = \left(\frac{I_n - \Pi_{\mathbb{1}} - \Pi_A}{\sigma^2} \right) \sigma^2 I_n = I_n - \Pi_{\mathbb{1}} - \Pi_A$ is idempotent

$\text{rank}(A) = \text{tr}(I_n - \Pi_{\mathbb{1}} - \Pi_A) = \text{tr}(I_n) - \text{tr}(\Pi_{\mathbb{1}}) - \text{tr}(\Pi_A) = n - 1 - 1 = n - 2$

$\delta = \mu^T A \mu = E[y]^T \left(\frac{I_n - \Pi_{\mathbb{1}} - \Pi_A}{\sigma^2} \right) E[y]$

$E[y] = \beta_0 \mathbb{1} + \beta_1 X$
 $= \beta_0 \mathbb{1} + \beta_1 \left(\frac{X - \bar{x} \mathbb{1}}{S_{xx}} \cdot S_{xx} + \bar{x} \mathbb{1} \right)$

$= \beta_0 \mathbb{1} + \beta_1 (A S_{xx} + \bar{x} \mathbb{1})$

$= (\beta_0 + \beta_1 \bar{x}) \mathbb{1} + \beta_1 S_{xx} A$

$\therefore$ linear combination of $\mathbb{1}$ and A

$E[y] \in \text{Span}(\mathbb{1}, A)$ and $(I_n - \Pi_{\mathbb{1}} - \Pi_A) \in \text{Span}(\mathbb{1}, A)^\perp$

$\therefore \delta = 0$

(c) $\hat{\beta}_1$ and $\hat{\sigma}^2$ are independent

$\hat{\beta}_1 \in C_A$ (column space of A)
 $\hat{\sigma}^2 \in \text{Span}(\mathbb{1}, A)^\perp$

$\therefore \hat{\beta}_1$ and $\hat{\sigma}^2$ are orthogonal

Remark: Notice that SSE and SSR are independent with

$$SSE/\sigma^2 \sim \chi^2(n-2) \quad SSR/\sigma^2 \sim \chi^2(1; \delta), \quad \delta = \frac{\beta_1^2}{\sigma^2/S_{xx}}$$

$$SSE/\sigma^2 = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \sim \chi^2(n-2) \text{ from Q12-(b)}$$

$$\begin{aligned} SSR/\sigma^2 &= \frac{1}{\sigma^2} \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \frac{1}{\sigma^2} \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y})^2 = \frac{1}{\sigma^2} \sum_{i=1}^n (\hat{\beta}_1 x_i - \bar{y})^2 \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \bar{x}) \hat{\beta}_1 = \frac{1}{\sigma^2} \sum_{i=1}^n \left\{ (x_i - \bar{x}) \cdot \frac{\sum_{j=1}^n (x_j - \bar{x}) y_j}{S_{xx}} \right\}^2 \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n \left\{ S_{xx} \cdot \frac{(x_i - \bar{x})}{S_{xx}} \cdot A^T y \right\}^2 \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n \left\{ \frac{(x_i - \bar{x})}{S_{xx}} \cdot (A^T A)^{-1} A^T y \right\}^2 \\ &= \frac{1}{\sigma^2} y^T A (A^T A)^{-1} A^T A (A^T A)^{-1} A^T y \\ &= \frac{1}{\sigma^2} y^T A (A^T A)^{-1} A^T y \\ &= \frac{1}{\sigma^2} y^T \Pi_A y \end{aligned}$$

$$\frac{1}{\sigma^2} y^T \Pi_A y \sim \chi^2(r; \delta), \quad r = \text{rank}(\Pi_A) = \text{rank}(A) = 1, \quad \delta = \mu^T A \mu = E[y]^T A E[y]$$

$$\begin{aligned} \delta &= (\beta_0 \mathbb{1} + \beta_1 x)^T \Pi_A (\beta_0 \mathbb{1} + \beta_1 x) / \sigma^2 \\ &= (\beta_1 x)^T \Pi_A (\beta_1 x) / \sigma^2 \quad \because A \text{ and } \mathbb{1} \text{ are orthogonal} \\ &= \beta_1^2 x^T A (A^T A)^{-1} A^T x / \sigma^2 \\ &= \beta_1^2 S_{xx} \mathbb{1}^T x^T A \mathbb{1}^2 / \sigma^2 \\ &= \beta_1^2 / \sigma^2 S_{xx} \left[\sum \frac{(x_i - \bar{x}) \cdot x_i}{S_{xx}} \right] \left[\sum \frac{(x_i - \bar{x}) x_i}{S_{xx}} \right] \\ &= \frac{\beta_1^2}{\sigma^2 S_{xx}} \quad \square \end{aligned}$$

HYPOTHESIS TESTING.

Q13)

$$H_0: \beta_1 = \beta_1^* \quad \text{vs} \quad H_1: \beta_1 \neq \beta_1^*$$

$$T = \frac{(\hat{\beta}_1 - \beta_1^*)}{\sqrt{\hat{\sigma}^2 / S_{xx}}}$$

(a) Show that under H_0 : $T \sim t(n-2)$

(b) Show that under H_1 : $T \sim t(n-2; \delta)$, $\delta = \frac{\beta_1 - \beta_1^*}{\sqrt{\sigma^2 / S_{xx}}}$

$$F = \frac{S_{xx}(\hat{\beta}_1 - \beta_1^*)^2}{SSE/(n-2)}$$

(c) Show that under H_0 : $F \sim F(1, n-2)$

(d) Show that under H_1 : $F \sim F(1, n-2; \delta)$, $\delta = \frac{(\beta_1^* - \beta_1)^2}{\sigma^2 / S_{xx}}$

$$(a) \quad T = \frac{(\hat{\beta}_1 - \beta_1^*)}{\sqrt{\hat{\sigma}^2 / S_{xx}}}, \quad \hat{\beta}_1 \sim N(\beta_1, \sigma^2 / S_{xx}) \Leftrightarrow \frac{\hat{\beta}_1 - \beta_1}{\sigma / \sqrt{S_{xx}}} \sim N(0, 1)$$

$$\therefore T = \frac{\hat{\beta}_1 - \beta_1^*}{\sigma / \sqrt{S_{xx}}} \sqrt{\frac{(n-2)\hat{\sigma}^2}{\sigma^2} / (n-2)} = \frac{Z}{\sqrt{V/(n-2)}}, \quad Z \sim N(0, 1) \text{ and } V \sim \chi^2(n-2)$$

$$\therefore T \sim t(n-2)$$

$$(b) \quad T = \frac{(\hat{\beta}_1 - \beta_1 + \beta_1 - \beta_1^*)}{\sigma / \sqrt{S_{xx}}} \sqrt{\frac{(n-2)\hat{\sigma}^2}{\sigma^2} / (n-2)} = \frac{Z + \delta}{\sqrt{V/(n-2)}}, \quad Z \sim N(0, 1), \quad V \sim \chi^2(n-2)$$

$$\therefore T \sim t(n-2; \delta), \quad \delta = \frac{\beta_1 - \beta_1^*}{\sigma / \sqrt{S_{xx}}}$$

$$(c) \quad F = \frac{S_{xx}(\hat{\beta}_1 - \beta_1^*)^2}{SSE/(n-2)}$$

$$\begin{aligned} \hat{\beta}_1 &\sim N(\beta_1, \sigma^2 / S_{xx}) \quad \therefore \frac{\hat{\beta}_1 - \beta_1}{\sigma / \sqrt{S_{xx}}} = \frac{\sqrt{S_{xx}}(\hat{\beta}_1 - \beta_1)}{\sigma} \sim N(0, 1) \\ &\therefore \frac{S_{xx}(\hat{\beta}_1 - \beta_1)^2}{\sigma^2} \sim \chi^2(1) \end{aligned}$$

$$\frac{SSE}{\sigma^2} = \frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-2)$$

$$\therefore F = \frac{V_1/1}{V_2/(n-2)} \sim F(1, n-2) \quad \because \hat{\beta}_1 \text{ and } \hat{\sigma}^2 \text{ are independent (i.e., } \hat{\beta}_1 \perp \hat{\sigma}^2) \\ V_1 \sim \chi^2(1) \text{ and } V_2 \sim \chi^2(n-2)$$

(d) under H_1 : $\beta_1 \neq \beta_1^*$,

$$F = \frac{(\hat{\beta}_1 - \beta_1^*)^2}{\sigma^2 / S_{xx}} \frac{(n-2)\hat{\sigma}^2}{\sigma^2} / (n-2), \quad \frac{\hat{\beta}_1 - \beta_1^*}{\sigma / \sqrt{S_{xx}}} = \frac{\hat{\beta}_1 - \beta_1}{\sigma / \sqrt{S_{xx}}} + \frac{\beta_1 - \beta_1^*}{\sigma / \sqrt{S_{xx}}} = Z + \delta \stackrel{d}{=} N(\sqrt{\delta}, 1)$$

$$\therefore F = \frac{(Z + \delta)^2}{V_2/(n-2)} = \frac{V_1^*/1}{V_2/(n-2)}, \quad V_1^* \sim \chi^2(1; \delta) \text{ and } V_2 \sim \chi^2(n-2)$$

$$\therefore F \sim \chi^2(1, n-2; \delta) \quad \square$$

Q15)

$$H_0: \beta_0 = \beta_0^* \text{ vs } H_1: \beta_0 \neq \beta_0^*$$

$$T = \frac{\hat{\beta}_0 - \beta_0^*}{\sqrt{\hat{\sigma}^2(n^{-1} + \bar{x}^2 S_{xx}^{-1})}}$$

(a) Show that under H_0 , $T \sim t(n-2)$ (b) Show that $H_1: T_1 \sim t(n-2; \delta)$, $\delta = \frac{\beta_0 - \beta_0^*}{\sqrt{\hat{\sigma}^2(n^{-1} + \bar{x}^2 S_{xx}^{-1})}}$

$$(a) \hat{\beta}_0 \sim N(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right))$$

$$(\hat{\beta}_0 - \beta_0) / \left(\sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \right)^{1/2} \sim N(0, 1)$$

$$\text{under } H_0: \beta_0 = \beta_0^*, (\hat{\beta}_0 - \beta_0^*) / (\sigma^2(n^{-1} + \bar{x}^2 S_{xx}^{-1})) \sim N(0, 1)$$

$$\begin{aligned} \therefore T &= \frac{\hat{\beta}_0 - \beta_0^*}{\sqrt{\hat{\sigma}^2(n^{-1} + \bar{x}^2 S_{xx}^{-1})}} = \frac{\hat{\beta}_0 - \beta_0^*}{\sqrt{\sigma^2(n^{-1} + \bar{x}^2 S_{xx}^{-1})}} \bigg/ \sqrt{\frac{(n-2)\hat{\sigma}^2}{\sigma^2} / (n-2)} \\ &= Z / \sqrt{V/(n-2)}, \quad Z \sim N(0, 1), \quad V \sim \chi^2(n-2) \\ &\quad \hat{\beta}_0 \text{ and } \hat{\sigma}^2 \text{ are independent.} \\ &\quad \therefore Z \text{ and } V \text{ are independent.} \end{aligned}$$

$$\therefore T = \frac{Z}{\sqrt{V/(n-2)}} \sim t(n-2)$$

$$\begin{aligned} (b) T &= \frac{\hat{\beta}_0 - \beta_0^*}{\sqrt{\hat{\sigma}^2(n^{-1} + \bar{x}^2 S_{xx}^{-1})}} = \frac{(\hat{\beta}_0 - \beta_0) + (\beta_0 - \beta_0^*)}{\sqrt{\hat{\sigma}^2(n^{-1} + \bar{x}^2 S_{xx}^{-1})}} \bigg/ \sqrt{\frac{(n-2)\hat{\sigma}^2}{\sigma^2} / (n-2)} \\ &= (Z + \delta) / \sqrt{V/(n-2)}, \quad Z \sim N(0, 1), \quad \delta = \frac{\beta_0 - \beta_0^*}{\sqrt{\sigma^2(n^{-1} + \bar{x}^2 S_{xx}^{-1})}} \\ &\quad V \sim \chi^2(n-2) \\ &\quad Z \perp V \end{aligned}$$

$$\therefore T \sim t(n-1; \delta)$$

Confidence Interval

Q16) Derive the (1- α)100% confidence interval for β_1 and β_0

$$\frac{\hat{\beta}_1 - \beta_1^*}{\hat{\sigma} / \sqrt{S_{xx}}} \sim t(n-2)$$

$$\therefore (1-\alpha)100\% \text{ CI is: } \left| \frac{\hat{\beta}_1 - \beta_1^*}{\hat{\sigma} / \sqrt{S_{xx}}} \right| < t_{\alpha/2}(n-2)$$

$$\Rightarrow \hat{\beta}_1 - \frac{\hat{\sigma}}{\sqrt{S_{xx}}} t_{\alpha/2}(n-2) < \beta_1^* < \hat{\beta}_1 + \frac{\hat{\sigma}}{\sqrt{S_{xx}}} t_{\alpha/2}(n-2)$$

$$(1-\alpha)100\% \text{ CI of } \beta^*: \hat{\beta}_1 \pm \frac{\hat{\sigma}}{\sqrt{S_{xx}}} t_{\alpha/2}(n-2)$$

$$\frac{\hat{\beta}_0 - \beta_0^*}{\sqrt{\hat{\sigma}^2(n^{-1} + \bar{x}^2 S_{xx}^{-1})}} \sim t(n-2)$$

$$\therefore (1-\alpha)100\% \text{ CI of } \beta_0^*: \hat{\beta}_0 \pm t_{\alpha/2}(n-2) \cdot \hat{\sigma} \sqrt{(n^{-1} + \bar{x}^2 S_{xx}^{-1})}$$

Q17) Prove the following:

$$(a) \mu_1 \text{ CI: } \hat{\mu}_1 \pm t_{\alpha/2}(n-2) \sqrt{\hat{\sigma}^2 \left(\frac{1}{n-1} + \frac{(x_1 - \bar{x})^2}{S_{xx}} \right)}$$

$$(b) y_2 \text{ CI: } \hat{y}_2 \pm t_{\alpha/2}(n-2) \sqrt{\hat{\sigma}^2 \left(1 + n^{-1} + \frac{(\bar{x} - \bar{x})^2}{S_{xx}} \right)}$$

(a) $\mu_1 = E[y_1] = \beta_0 + \beta_1 x_1$

$$\hat{y}_1 = \hat{\beta}_0 + \hat{\beta}_1 x_1 = \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_1 = \bar{y} + \hat{\beta}_1 (x_1 - \bar{x})$$

$$\bar{y} \sim N(\mu_1, \sigma_n^2), \quad \hat{\beta}_1 \sim N(0, \frac{\sigma^2}{S_{xx}}), \quad \text{cov}(\bar{y}, \hat{\beta}_1) = 0$$

$$\begin{aligned} \therefore \bar{y} + (x_1 - \bar{x}) \hat{\beta}_1 &\sim N(\mu_1, \text{Var}(\bar{y}) + \text{Var}(\hat{\beta}_1) \cdot (x_1 - \bar{x})^2) \\ &\sim N(\mu_1, \sigma^2(n^{-1} + \frac{(x_1 - \bar{x})^2}{S_{xx}})) \end{aligned}$$

$$\therefore (\hat{\mu}_1 - \mu_1) / \sqrt{\sigma^2(n^{-1} + \frac{(x_1 - \bar{x})^2}{S_{xx}})} \sim N(0, 1)$$

$$(\hat{\mu}_1 - \mu_1) / \sqrt{\hat{\sigma}^2(n^{-1} + \frac{(x_1 - \bar{x})^2}{S_{xx}})} \sim t(n-2) \quad \therefore \hat{\mu}_1 \text{ and } \hat{\sigma}^2 \text{ are independent}$$

$$\hat{\mu}_1 = \hat{\beta}_0 + \hat{\beta}_1 x_1, \quad \hat{\beta}_0 = \bar{y} + \hat{\beta}_1 \bar{x} = (\mathbf{1}^T \mathbf{1})^{-1} \mathbf{1}^T \mathbf{y} + \bar{x} (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y} \in \text{Span}(\mathbf{1}, \mathbf{A})$$

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n-2} \mathbf{y}^T (\mathbf{I}_n - \Pi_{\mathbf{1}} - \Pi_{\mathbf{A}}) \mathbf{y} \in \text{Span}(\mathbf{1}, \mathbf{A})^\perp$$

 $\therefore \hat{\mu}_1$ and $\hat{\sigma}^2$ are independent

$$(1-\alpha)100\% \text{ CI of } \mu_1: \hat{\mu}_1 \pm t_{\alpha/2}(n-2) \left[\hat{\sigma}^2 \left(n^{-1} + \frac{(x_1 - \bar{x})^2}{S_{xx}} \right) \right]$$

(b) $y_2 = \beta_0 + \beta_1 z + \epsilon_z, \quad \epsilon_z \sim N(0, \sigma^2)$

$$\hat{y}_2 = \hat{\beta}_0 + \hat{\beta}_1 z = \bar{y} + \hat{\beta}_1 (z - \bar{x}) \sim N(\beta_0 + \beta_1 z, \sigma^2(n^{-1} + \frac{(z - \bar{x})^2}{S_{xx}}))$$

$$y_2 \sim N(\beta_0 + \beta_1 z, \sigma^2)$$

$$\therefore \hat{y}_2 - y_2 \sim N(0, \sigma^2(1 + n^{-1} + \frac{(z - \bar{x})^2}{S_{xx}}))$$

$$\frac{\hat{y}_2 - y_2}{\left[\sigma^2(1 + n^{-1} + \frac{(z - \bar{x})^2}{S_{xx}}) \right]^{1/2}} \sim N(0, 1)$$

$$\therefore \frac{\hat{y}_2 - y_2}{\left[\hat{\sigma}^2(1 + n^{-1} + \frac{(z - \bar{x})^2}{S_{xx}}) \right]^{1/2}} \sim t(n-2) \quad \therefore \hat{y}_2 \text{ and } \hat{\sigma}^2 \text{ are independent.}$$

$$\therefore (1-\alpha)100\% \text{ CI of } y_2: \hat{y}_2 \pm t_{\alpha/2}(n-2) \left[\hat{\sigma}^2 \left(1 + n^{-1} + \frac{(z - \bar{x})^2}{S_{xx}} \right) \right]^{1/2}$$

Q18) Prove the following:

(a) $\hat{\mu}_1$ and $\hat{\sigma}^2$ are independent

(b) $\hat{y}_2$ and $\hat{\sigma}^2$ are independent

(a) $\hat{\mu}_1 = \hat{\beta}_0 + \hat{\beta}_1 x_i$. We know that $\hat{\beta}_0, \hat{\beta}_1$ are independent to $\hat{\sigma}^2$ (proof 7)

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = (S_{xx})^{-1} \sum (x_i - \bar{x}) y_i = (A^T A)^{-1} A^T y, \quad A = [x_1 - \bar{x}, \dots, x_n - \bar{x}]^T, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \bar{y} - \hat{\beta}_1 (x_i - \bar{x}))^2$$

$$= \frac{1}{n-2} \sum_{i=1}^n (y_i - (\mathbf{1}^T \mathbf{1})^{-1} \mathbf{1}^T y - (A^T A)^{-1} A^T y (x_i - \bar{x}))^2$$

$$= \frac{1}{n-2} (y - \Pi_{\mathbf{1}} y - \Pi_A y)^2 = (n-2)^{-1} y^T (\mathbf{I}_n - \Pi_{\mathbf{1}} - \Pi_A) y \in \text{Span}(\mathbf{1}, A)^\perp$$

$\therefore \hat{\beta}_1$ and $\hat{\sigma}^2$ are independent

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = (\mathbf{1}^T \mathbf{1})^{-1} \mathbf{1}^T y - \hat{\beta}_1 \bar{x} \in \text{Span}(\mathbf{1}, A)$$

$\therefore \hat{\beta}_0$ and $\hat{\sigma}^2$ are independent

$\therefore \hat{\mu}_1$, which is a function of $\hat{\beta}_0$ and $\hat{\beta}_1$, is also independent to $\hat{\sigma}^2$

(b) $\hat{y}_2 = \hat{\beta}_0 + \hat{\beta}_1 z$ is a function of $\hat{\beta}_0$ and $\hat{\beta}_1$

$\therefore \hat{y}_2$ is independent to $\hat{\sigma}^2$ since $\hat{\beta}_0, \hat{\beta}_1$ are independent to $\hat{\sigma}^2$.